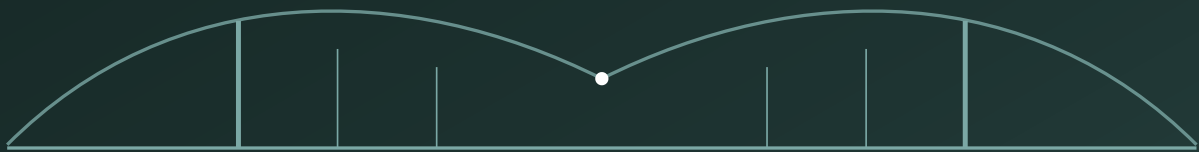


User Guide

Find a structure, run an inspection, read the BCI score, and get the report out. Everything you need to use spanSense day to day, walked through step by step with real screenshots of the current platform.



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A map of this guide.

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01. Getting Started

A quick orientation before your first inspection.

Log in with the account your admin gave you, and you'll land on the Map. Its top bar just holds the logo, a search box, the night-mode toggle, your account icon, and Contact Us; the links to **Dashboard**, **Database**, **Planning**, and **twinView** live in a collapsible sidebar on the left edge instead (hover it to expand). Once you move to any of those other pages, that same set of links moves up into a floating navbar at the top. The account icon's shape tells you your role at a glance: a magnifying glass for inspectors, a checked user for engineers, a shield for admins.

Your role decides what you can do

ROLE	CAN DO
Inspector	Find structures, run inspections, upload photos, view reports and planning
Engineer	Everything an inspector can, plus approve/reject submitted inspections and edit a structure's inspection schedule
Admin	Everything an engineer can, plus full system access

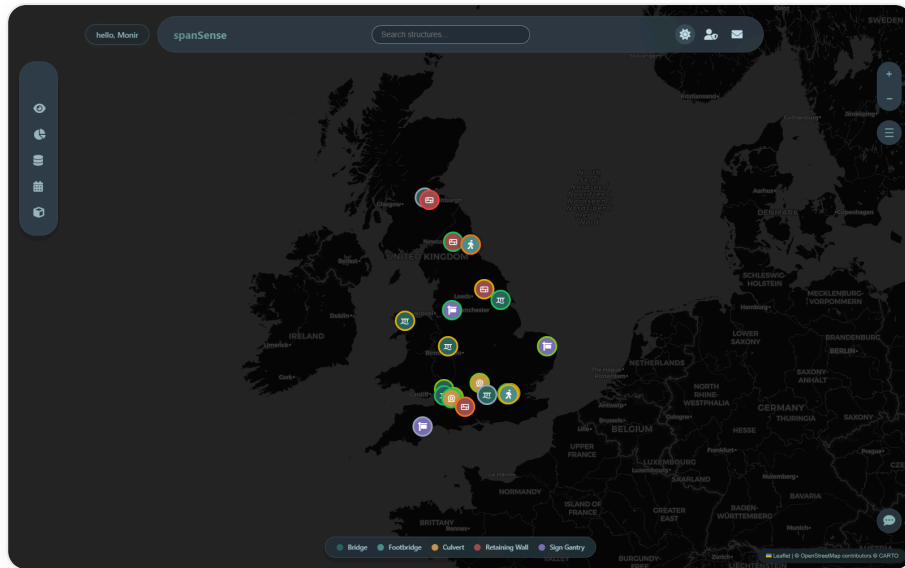
The general workflow

- 1 **Find** your structure on the Map
- 2 **Run** a new inspection, or open an existing one from Previous Inspections
- 3 **Save** it, and the BCI score is calculated automatically as you go
- 4 An **engineer reviews** it from the Dashboard's Pending Review queue
- 5 **Export** a report whenever you need one, from the structure card or Database

Tip: you don't need to finish an inspection in one sitting; every wizard step has a **Save Draft** option that picks up right where you left off.

02. The Map: Finding a Structure

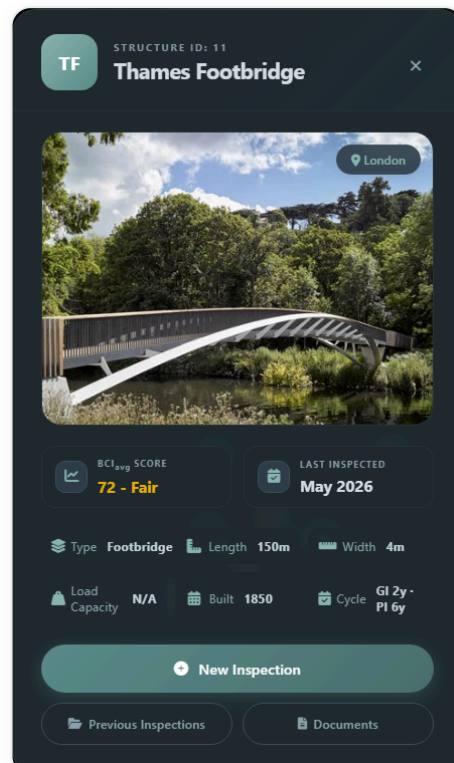
The map is your home screen after login, and the starting point for almost everything.



Every structure plotted by type icon and condition-coded ring

Structures are plotted with a type-coded icon (bridge, footbridge, culvert, retaining wall, or sign gantry) and a condition-coded ring color, the same colors used everywhere else in the app, so a glance at the map already tells you which structures need attention.

- 1 Use the **search bar** at the top to jump straight to a named structure, or the sidebar filters on the left to narrow by type or condition
- 2 **Click a marker** to open its structure card: photo, BCI average score, last inspected date, span/length/year/type
- 3 From the card, choose **New Inspection**, **Previous Inspections**, or **Documents**



Click a marker to open its structure card

03. Viewing Previous Inspections

Every past inspection for a structure, in one searchable list.

Previous Inspections (opened from the structure card) shows the full record for that structure: total inspection count, the last inspection date, and the current BCI average with its trend against the one before it.

- Filter by **Principal (PI)**, **General (GI)**, or **Safety (SI)**
- Search across every recorded inspection for that structure
- Each row shows the inspector, date, and both **CRIT** and **AV** scores
- Use the row's icons to download the **PDF report**, **edit** the inspection, or generate a **BCI Proforma**
- Click **+ New Inspection** to start the next one straight from here

IB

PREVIOUS INSPECTIONS

Iron Bridge

Structure ID: 14

TOTAL
23

LAST INSPECTION
Nov 2026

CURRENT BCI_{avg}
70 ↓ 3

✓ All

Principal (PI)

General (GI)

Safety (SI)

Search inspections...

2026
NOV 7

PRINCIPAL

Monir Khan

CRIT
33

AV
70



2026
JUL 23

PRINCIPAL

Monir Khan

CRIT
0

AV
73



Click any inspection to view full details

Close

+ New Inspection

Filter, search, and act on any past inspection for this structure

04. Running an Inspection, Step by Step

The wizard is two steps: metadata, then defect logging element by element.

Step 1: Metadata

Confirm the inspection type (GI/PI/SI), date, and inspector, then work through each span's above-ground/photograph checks before moving on to the element checklist.

Step 2: Log each element

- 1 Every inspectable element is listed as a row. Click one and choose **Add Defect**, **No Defects**, or **Not Inspected** (with a reason)
- 2 For a defect, set **Severity** (1 Minor → 5 Emergency) and **Extent** (A isolated → E extensive). A live preview shows the BCI impact as you adjust them, before you save
- 3 Mark **Works Required** (No, Yes, or Monitor) and attach a photo if useful
- 4 If a rating is a big improvement over what was last recorded for that element, a confirmation prompt double-checks it before saving, as a quick safeguard against typos
- 5 **Save** the defect and move to the next row, or **Save Draft** at any point to finish later

S Severity 1–5
Minor · Moderate · Severe · Critical · Emergency

E Extent A–E
How much of the element is affected, isolated to extensive

W Works Required
No, Yes, or Monitor; flags it for the planning programme

✓ Conclusions
A short written summary once every element is done

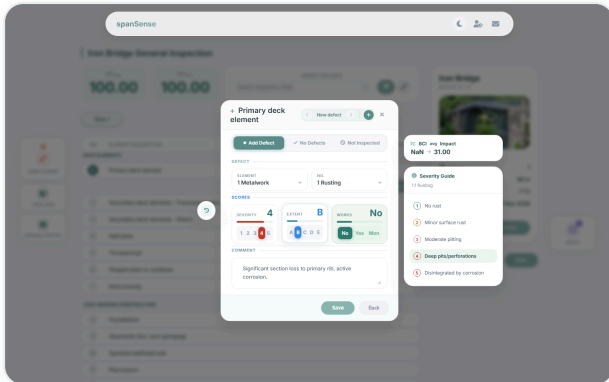
Finishing up

Once every element is logged, add your **Conclusions** and hit **Save**. The BCI average and critical scores are calculated automatically from what you've logged, no manual math required. The inspection then enters the review queue for an engineer to approve.

General Photos: not every useful photo belongs to one specific element. The **General Photos** bar in the left rail (next to Conclusions and twinView) holds site shots that don't fit any single defect, for sorting out later.

04. Inside the Add Defect Modal

A closer look at the previous-defects panel, the live BCI preview, the severity guide, and moving between defects, all without leaving the modal.



Previous Defects, and copying one in

Open a defect for an element that's had one logged before, and a **Previous Defects** panel appears beside the modal, collapsed to a slim tab until you expand it. Each past defect shows as a card (Severity/Extent/Works badges, date, comment), three most recent first with a **Load more** option for the rest. Click a card to copy it straight into the form: severity, extent, works, cost, remedial works, and comment all fill in together, with a **"From <date>"** badge confirming it's a copy. Adjust what's changed, then save.

The Add Defect modal in night mode: Previous Defects expanded on the left with real history, Severity Guide flyout on the right

Not the same as the table's Copy icon: the Copy/Edit icons on an already-logged row duplicate a defect you've just saved in *this* inspection; Previous Defects pulls from history instead.

Live BCI Impact preview

Move the Severity or Extent stepper and the **BCI_avg Impact** panel recalculates instantly: current score on the left, what it would become on the right. It uses the same calculation that produces the final report score, so there's no surprise once you save: red means your change makes the structure's score worse, green means it improves.



Severity Guide

Not sure what separates a 3 from a 4? This panel spells out all five levels in plain language for the exact defect type and number picked, with your current selection highlighted.

LVL	1.1 RUSTING
1	No rust
2	Minor surface rust
3	Moderate pitting
4	Deep pits/perforations
5	Disintegrated by corrosion

Moving between defects

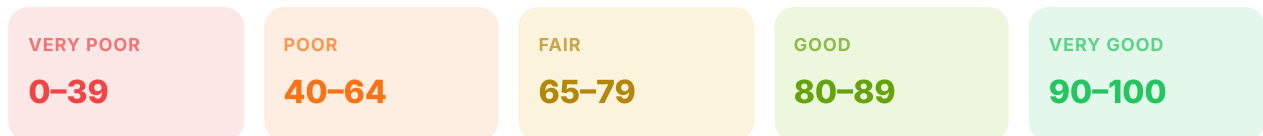
No need to close the modal to work through an element's defects:

- Use the ◀/▶ arrows in the modal header to step between defects already logged for this element; a **"Defect X of N"** counter shows where you are, and each move auto-saves what you were editing
- Click + to start logging another new defect on the same element without closing the modal

05. Understanding BCI Scores

BCI_{av} and BCI_{crit} are calculated for you: no formula, no spreadsheet.

BCI_{av} is the condition across every inspected element, weighted by how structurally important each one is. **BCI_{crit}** is narrower: each structure type has a fixed set of **critical elements** (e.g. main girders, bearings, abutments) and BCI_{crit} is based purely on the worst condition among just those, not the worst element in the whole inspection. A bad defect on a non-critical element (a sign, a handrail) drags BCI_{av} down but leaves BCI_{crit} untouched; if none of the critical elements have a defect, BCI_{crit} reads 100. Both fall into the same five bands used everywhere in the app:



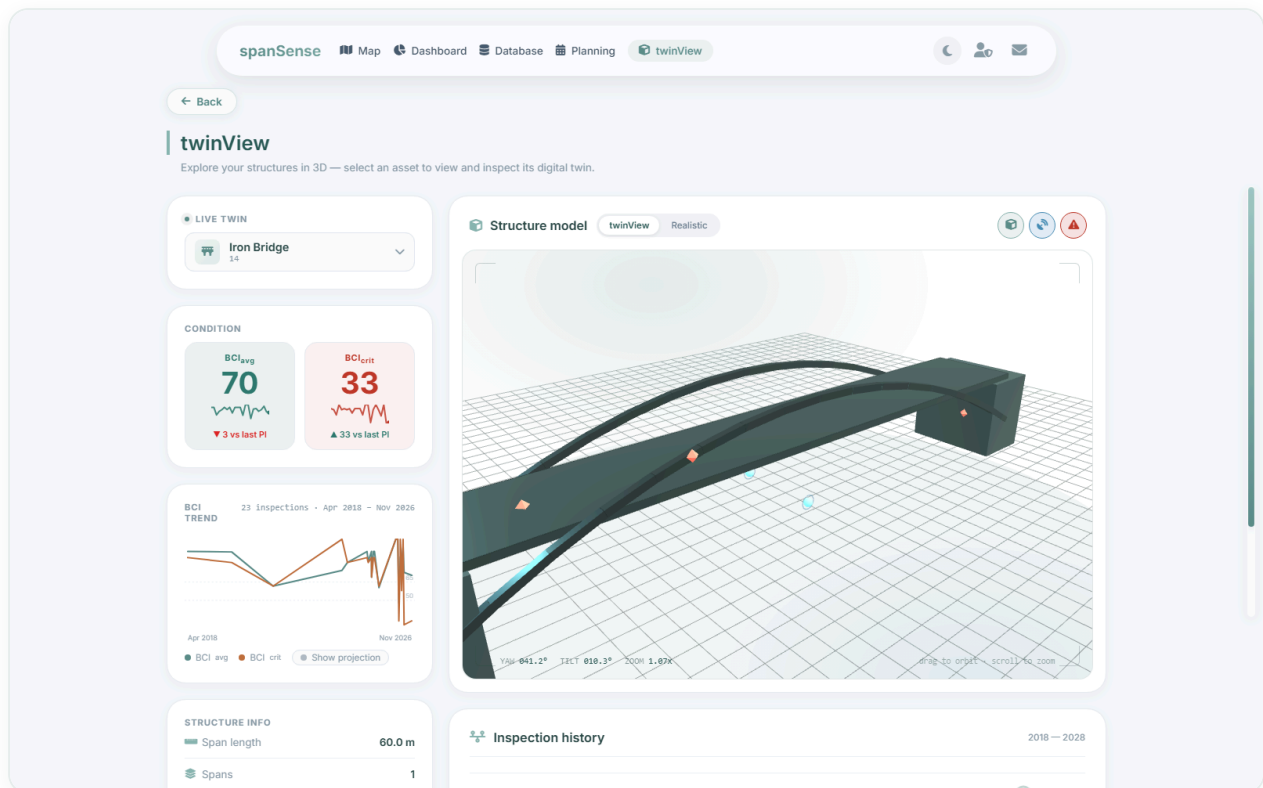
You'll see this exact banding in three places: the structure card, the Add Defect modal's live impact preview, and the Dashboard's BCI Score Distribution chart. twinView's condition panel is a bit coarser: it colors the score as just Critical, Fair, or Good, so treat it as a quick at-a-glance signal rather than the precise five-band read.

Where to find it: the structure card shows the current average; Previous Inspections shows CRIT and AV per inspection; twinView shows both, plus how they've changed since the last inspection.

06. twinView: 3D Digital Twin

A 3D model of the structure, built from real span and condition data.

- 1 Pick a structure from the **Live Twin** selector at the top left
- 2 **Drag to orbit**, scroll to zoom, and toggle **Structure**, **Sensors**, **Defects**, and **Works Required** layers on or off
- 3 The **Condition** card shows current BCI avg/crit; **Inspection Status** below it shows the trend against the previous inspection
- 4 Use the **Inspection History** timeline to jump to any past inspection and see the model as it was then



Orbit and zoom the 3D model; the condition trend is right there in Inspection Status

07. Reports & Exporting Data

Generate a report from one inspection, or export your whole stock from Database.

- **Inspection PDF / BCI Proforma:** generated straight from a saved inspection, in Previous Inspections or the inspection itself
- **Structure records:** CSV, JSON, or XML from Database, filterable by structure type
- **Inspection records:** CSV, PDF, or XLSX, with a column picker so you only export what you need
- **Report files:** PDF or DOCX summary documents

spanSense | Map | Dashboard | **Database** | Planning | twinView

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Database Export

Export bridge inspection data and reports in multiple formats

Bridges
Export all bridge records including structure details, location, and metadata
24 records | CSV, JSON, XML

Inspections
Export inspection history with dates, inspectors, and condition scores
133 records | CSV, PDF, XLSX

Reports
Export generated inspection reports and summary documents
133 records | BCI, DOCX, PDF

Inspection Records

✓ All Inspections | General (GI) | Principal (PI) | Safety (SI) | Field | Desktop

INSPECTION ID	STRUCTURE ID	BRIDGE	INSPECTOR	DATE	BCI _{avg}	BCI _{crit}	TYPE	TOTAL SPANS	SOURCE
141	22	A47 Norwich Southern Bypass Gantry	Monir Khan	24 Jul 2026	86	100	General	1	Field
137	24	M60 Barton Gantry	Monir Khan	15 Jul 2026	100	100	General	1	Desktop
135	14	Iron Bridge	Monir Khan	23 Jul 2026	73	0	Principal	1	Field
134	7	Menai Suspension Bridge	Monir Khan	22 Jul 2026	66	66		2	Field

Choose a category, pick a format, and export; filters and a column picker are below

Document Status page

Every PDF and Word report opens with a **Document Status** page: who prepared the inspection, its current review status, who reviewed it and when, and their comments if any, straight from the review queue. It's the front-matter a formal document needs, generated automatically, with no extra step for you.

Document Status

Current review state of this inspection record, at the point this report was generated.

Approved — reviewed 22 Jul 2026 by Priya Nathan

Structure	Iron Bridge (14)
Inspection date	23 Jul 2026
Inspection type	Principal Inspection
Prepared by	Tom Ferris Field
Reviewed by	Priya Nathan
Reviewed on	22 Jul 2026
Status	Approved

REVIEWER COMMENTS

"BCI critical figure cross-checked against site photos - consistent with the recorded rib corrosion. Cleared for issue without amendment."

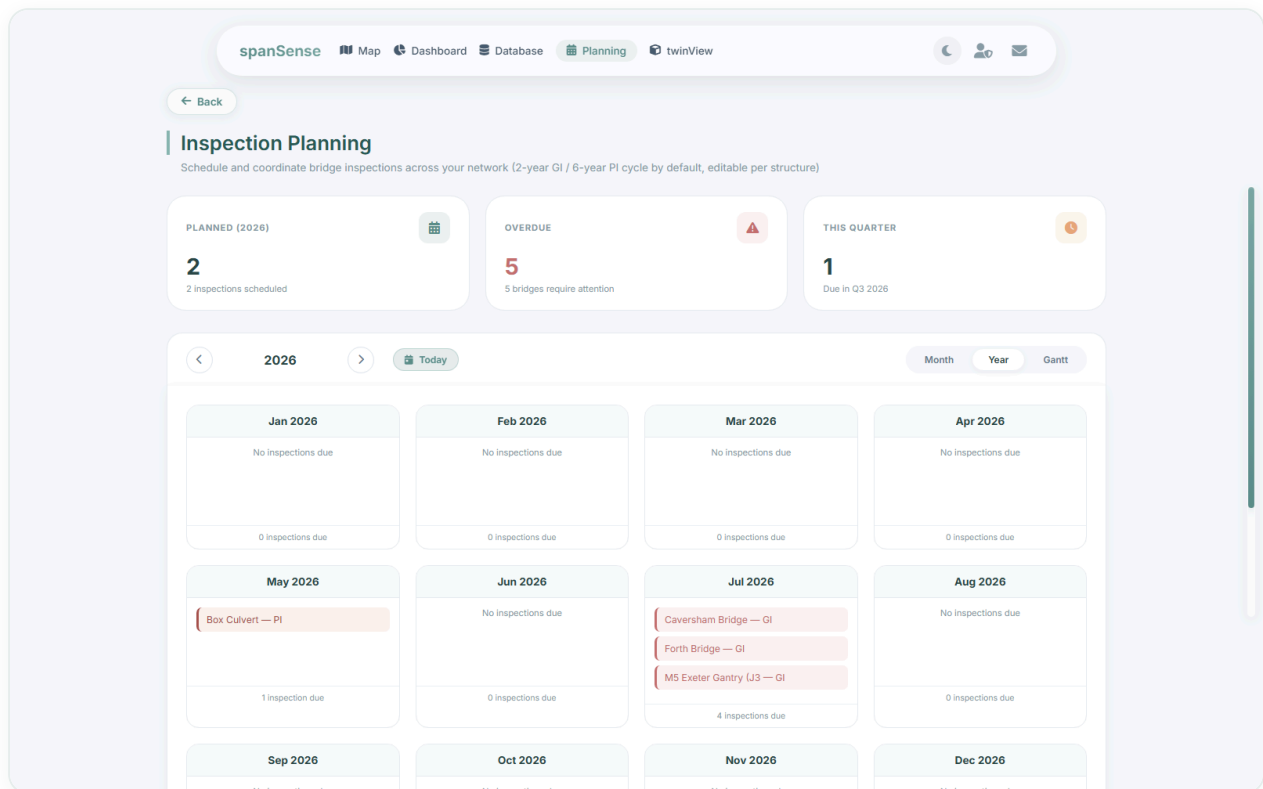
08. Inspection Planning & Scheduling

Never miss the next inspection, and it's tailored per structure.

By default, every structure follows a 2-year General Inspection / 6-year Principal Inspection cycle, alternating GI, GI, PI. Planning shows this as a **Month** calendar, a **Year** grid, or a multi-year **Gantt** view; click any inspection chip in Month or Year view to open its schedule editor directly.

Editing a structure's schedule

- 1 Click the **pencil icon** next to any structure in the Gantt view (or an inspection chip in Month/Year view)
- 2 Set a custom **GI** or **PI frequency** in years if this structure needs a different cycle than the default
- 3 Or set a **next inspection due** date directly, if you need to move just the next one without changing the ongoing cycle
- 4 **Save:** every calendar view updates immediately

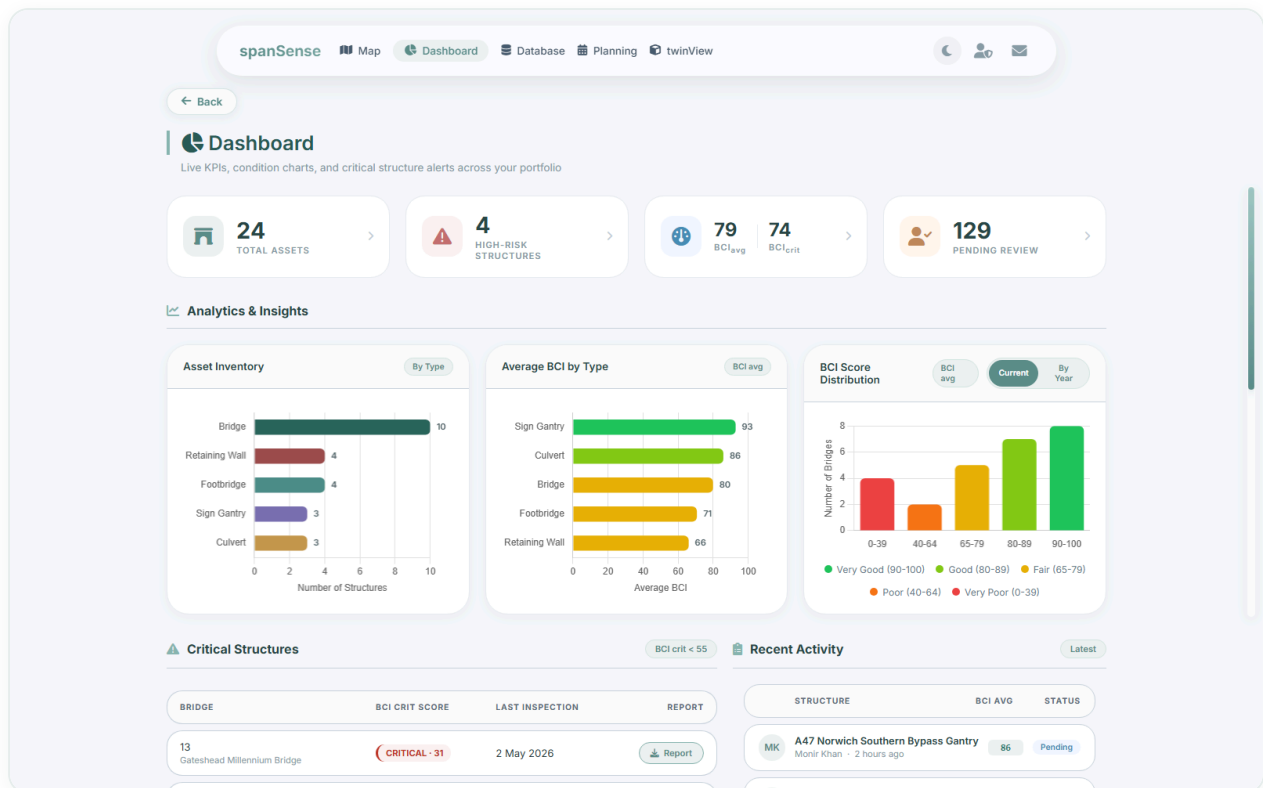


Year view: overdue inspections in red, upcoming in teal, principal inspections stand out separately

09. Dashboard: Portfolio Overview

Your whole stock at a glance, every card is clickable.

- **Total Assets** and **High-Risk Structures**: live counts, click to jump to the relevant chart or table
- **BCI Avg / Crit**: portfolio-wide condition, computed from every structure's latest inspection
- **Pending Review**: engineers and admins only; click to jump straight to the review queue
- **Asset Inventory**, **Average BCI by Type**, and **BCI Score Distribution** charts below



Four live metric cards and the Analytics & Insights charts below

If you're an engineer or admin

The **Pending Review** section further down lists every inspection awaiting a decision. Search, filter by type, or toggle "Critical only," then click **Review** to approve or reject with a comment, or open the full inspection first if you want to check the detail before deciding.

10. Your Account & Preferences

Open Account from any navbar to update your details or preferences.

**Profile Details**
Name, email, and organisation details

**Security**
Change your password from here

**Night Mode**
Toggle from any navbar; your choice is remembered

**Greeting**
The Map greets you by name once you're signed in

spanSense

MapDashboardDatabasePlanningtwinView

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Account

Manage your profile, security settings, and preferences

MK

Monir Khan

Admin · joined January 2026

AdminVerified

Sign Out

Profile Details

First Name

Monir

Last Name

Khan

Email Address

monir@spansense.io

Phone Number

Organisation

spanSense

Job Title

Admin

Save Changes

Security

Password

Last changed 6 days ago

Current password

New password

Confirm new password

Update Password

Two-Factor Authentication

Google Authenticator, Microsoft Authenticator, or any TOTP app

Not enabled

Enable 2FA

Last Signed In

24 July 2026 at 22:06

Your real profile, role, and join date, loaded from your account, not a placeholder

Stuck on something this guide didn't cover? Use **Contact Us** from any navbar and someone will get back to you.

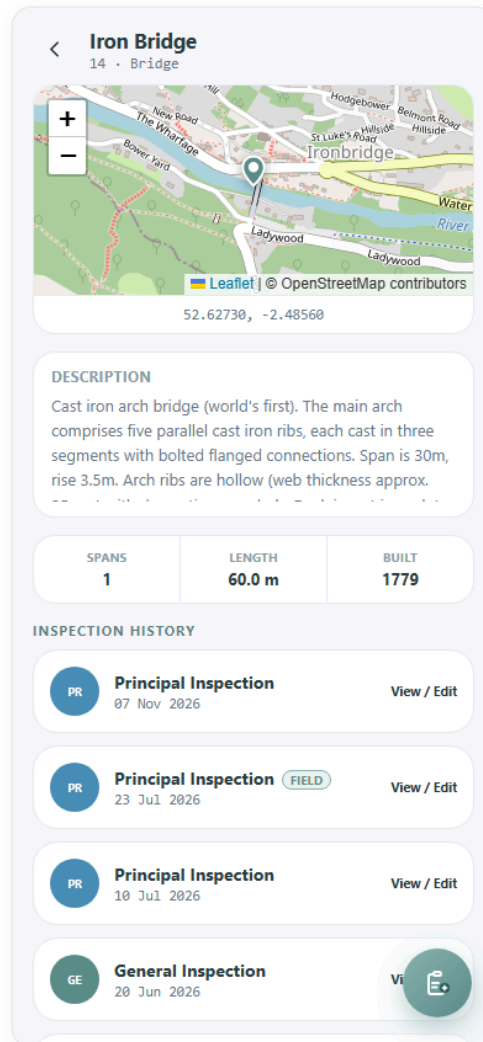
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11. spanSense Field: Your On-Site Companion

The same account, redesigned for standing at the bridge instead of sitting at a desk.

spanSense Field is a companion web app built for the phone in your pocket on site, not a separate product with its own login: sign in with the same spanSense account, and it works whether you're an inspector capturing a rough first pass or checking a structure's history before you even get out of the van. Install it to your home screen like a native app, and it keeps working with patchy or no signal, on scaffolding or under a deck, queuing your work and syncing automatically once you're back in range.



Every structure's own page: an interactive map, its description, and full inspection history

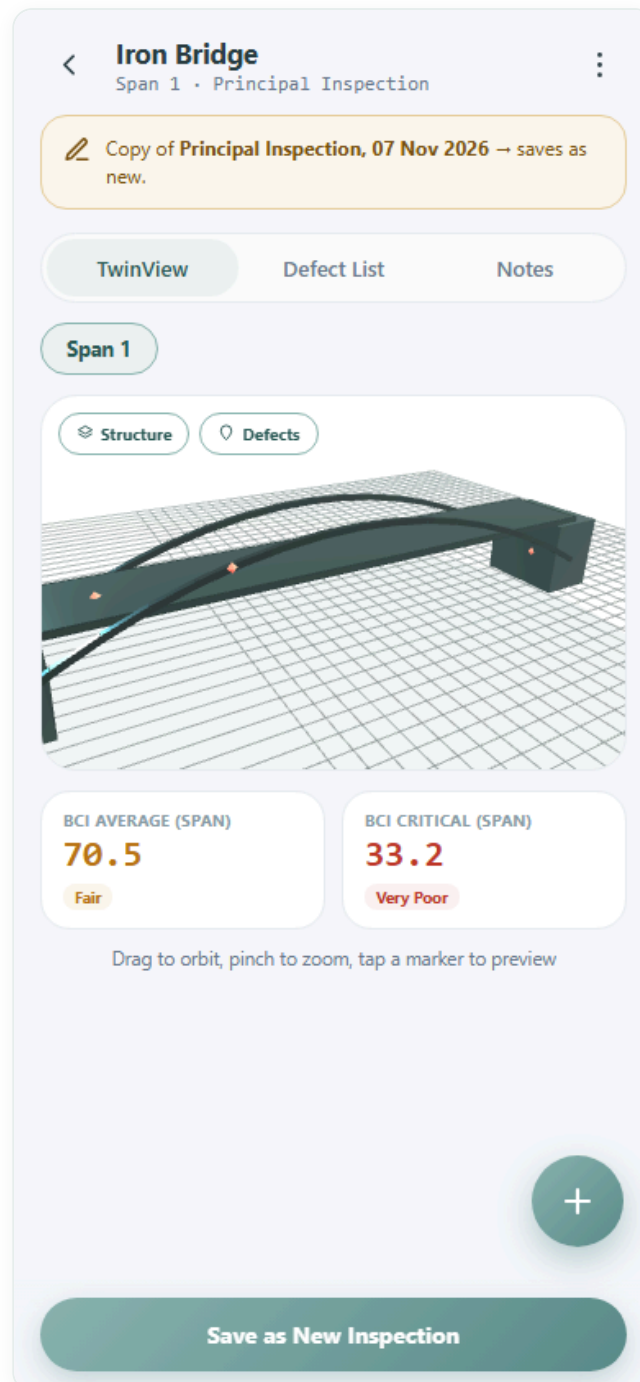
Starting an inspection on site

- 1 Search or scroll the **Structure List** to find the bridge you're standing at
- 2 Its page shows the location map, description, and every past inspection - tap the **+** to start a fresh one, or tap a past inspection to start a new draft copied from it, ready to correct and update
- 3 Log defects element by element exactly like spanSense: the same **Severity 1–5 / Extent A–E** system, so scores are consistent whichever app captured them

11. TwinView and Logging a Defect

The same 3D model as spanSense, in your hand at the structure.

TwinView on the phone is the same digital twin you'd get behind a desk: the model is built from the structure's real span and condition data, colored by defect severity, and orbits under your thumb instead of a mouse. Pinch to zoom in on a specific span, and tap any marker to preview the defect logged there without leaving the 3D view.



Drag to orbit, pinch to zoom, tap a marker to preview - BCI average and critical scored live for the current span

Why it's useful on site: compare the model against what you're looking at right now, rather than trying to remember it back at a desk later.

11. Logging a Defect

The same severity/extent system as spanSense, quicker to fill in with your thumb.

The screenshot shows the 'Edit defect' form in the spanSense application. The form is organized into several sections:

- Header:** A back arrow, the title 'Edit defect', a 'Span 1' label, and a 'Remove' button.
- ELEMENT:** A text field containing '2. Secondary deck elements - Transverse beams'.
- DEFECT TYPE:** A section with two small input fields containing '3' and '1', followed by a dropdown menu currently showing 'Deformation'.
- SEVERITY:** A row of five buttons labeled 1, 2, 3, 4, and 5. Button 4 is selected and highlighted in green.
- EXTENT:** A row of five buttons labeled A, B, C, D, and E. Button B is selected and highlighted in green.
- WORKS REQUIRED:** A row of three buttons labeled No, Yes, and Monitor. The 'No' button is selected and highlighted in green.
- COMMENTS:** A text area with the placeholder text 'defect 1.' and a microphone icon for voice recording.
- REMEDIAL WORKS:** A text area with the placeholder text 'What work would resolve this?' and a microphone icon for voice recording.
- Footer:** Two buttons at the bottom: 'Add Photo' with a camera icon and a large green 'Done' button.

The **Defect List** tab groups every element by category, with quick filter chips (Critical, Poor, Fair...) to jump straight to what needs attention.

Tap a defect to open the same editor as spanSense: element, defect type, severity, extent, and works required, with a photo attached directly to that defect.

Typing on site is slow with gloves on or a phone propped against a parapet, so **Comments** and **Remedial Works** both carry a microphone button for voice dictation - the same feature as the Notes tab, covered next.

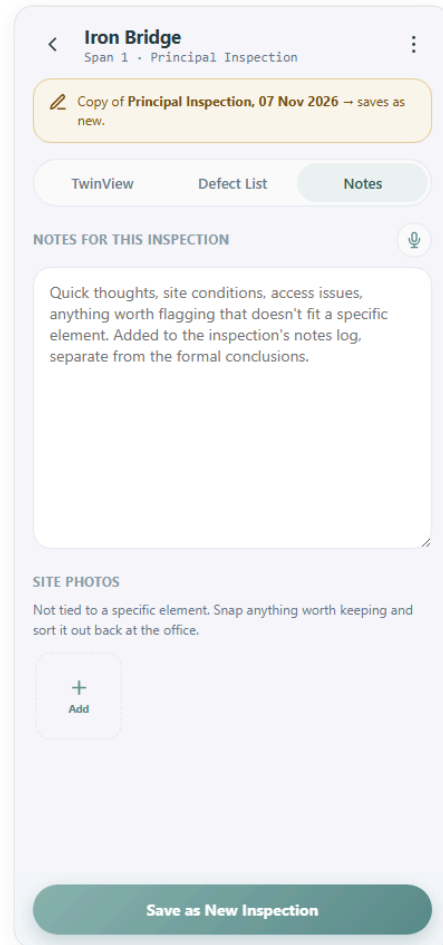
11. Notes, General Photos, and Staying in Sync

Dictate a note instead of typing it, and nothing is lost when you lose signal.

Notes & General Photos

The **Notes** tab is for anything that doesn't belong to one specific element: site conditions, access issues, a quick thought worth flagging before you forget it. It's a free-text field with the same voice dictation mic, saved onto the inspection's conclusions.

Right below it, **Site Photos** holds pictures that aren't tied to a defect either, the same feature as the General Photos bar in spanSense - snap it now, sort it out back at the office.



Working offline, and who sees it after

- **Offline-first:** lose signal mid-inspection and your save (photos included) queues on the device; a banner tells you you're offline, and everything syncs automatically the moment you're back online
- **Same review queue:** every Field-captured inspection is tagged with its source and flows into exactly the same engineer review queue as one authored in spanSense - nothing needs re-entering
- **Finished back at the office:** a Field draft is deliberately quick to capture rather than exhaustive; office staff can open it straight in spanSense's inspection editor afterwards (Database → Edit) to check it over or fill in anything that needed a second look

Tip: dark mode, your name, and your role carry over automatically - it's the same account and the same preferences, just a different screen.